LIME and SHAP: Explaining Machine-Learning Predictions

Seminar S04 Report – CS420

Nguyễn Trọng Hoàng (and team)

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Abstract

A highly accurate prediction does not always come with a reliable explanation: on Breast Cancer Wisconsin, two models reach 98.2% (Logistic Regression) and 95.6% (Random Forest) test accuracy, yet LIME and SHAP agree only on the easy case. This report explains the two post-hoc methods, works through hand-computed examples, and compares them on three instances chosen by a fixed rule. On the misclassified instance the top-5 features of LIME and SHAP overlap in only 2 of 5; LIME ranks *worst concave points* first with a negative sign, while SHAP's top eight features are all positive. The stability of the top-5 across seeds (LIME) or background sets (SHAP) ranges from 0.53 to 1.00 depending on the instance, and neither method is clearly more stable. A deletion test shows both are more faithful than random removal, and equal to each other. The conclusions hold only for one dataset, three instances and two models.

1 Introduction

Many modern machine-learning models are "black boxes": we see the input and the output but not which features drive a decision. In healthcare, finance or autonomous driving, a prediction cannot be trusted just because it is usually right, so the need to explain individual decisions keeps growing (Ribeiro et al., 2016; Mersha et al., 2024). LIME and SHAP are two widely used post-hoc methods for this purpose.

Guiding question. Does a highly accurate prediction necessarily come with a reliable explanation? To answer it, this report looks at three criteria: whether an explanation is useful (it shows which features push the prediction up or down), stable (it does not change with the seed or the background data) and faithful to the model (removing features called important should change the prediction accordingly).

Scope. We use the Breast Cancer Wisconsin dataset (569 instances, 30 features, two classes), two models (Logistic Regression and Random Forest) and three instances chosen by a fixed rule. The two source papers are Ribeiro et al. (2016) for LIME and Lundberg & Lee (2017) for SHAP; the survey by Mersha et al. (2024) is used to place both methods in the XAI taxonomy and to choose evaluation criteria.

Contributions.

- A synthesis and comparison of three sources: two required papers and one survey.
- Two hand-worked examples: SHAP on a linear model and LIME step by step on a small non-linear model.
- An experiment comparing top-k overlap, runtime and stability across seeds and background sets.
- A deletion test and a proposed procedure for evaluating faithfulness.

What belongs to the papers carries a citation; the remaining examples and numbers are our own.

2 Background and Paper Review

2.1 Core concepts

According to Mersha et al. (2024), terminology in XAI is not settled. We use the common distinction: *interpretability* is how well a human understands *how a model works*, while *explainability* is the ability to give reasons for *one specific decision*. Some sources treat one as a subset of the other, so the definition in use must be stated.

Table 1: Core concept pairs and where LIME and SHAP fall.

Concept pair	**Distinction**	**LIME and SHAP**
Explainability – interpretability	Explaining one prediction vs. understanding how the model works overall	Provide explainability; do not make the model interpretable
Local – global	Explaining one instance vs. describing the whole model	Both are local; LIME adds SP-LIME to pick representative instances for a global view
Model-specific – model-agnostic	Depends on model structure vs. needs only the prediction function	LIME and KernelSHAP are model-agnostic; Linear SHAP and Deep SHAP are model-specific
Intrinsic (ante-hoc) – post-hoc	Model is understandable by design vs. explained after training	Both are post-hoc and perturbation-based (Mersha et al., 2024)

2.2 LIME (Ribeiro et al., 2016)

LIME explains one prediction by generating perturbed samples around x, querying the black-box model on each, and fitting a simple surrogate model close to x. The explanation is:

$$\xi(x) = \arg\min_{g \in G} \; L(f, g, \pi_x) + \Omega(g) \tag{1}$$

$$L(f, g, \pi_x) = \sum_{z, z'} \pi_x(z)\big(f(z) - g(z')\big)^2, \qquad \pi_x(z) = \exp\big(-D(x, z)^2/\sigma^2\big) \tag{2}$$

Symbols: f is the black-box model (here $f(x)$ is the probability of one class, in $[0, 1]$); x is the instance to explain; z is a perturbed sample in the original space and z' its interpretable representation (a binary vector marking which components are present); G is the family of interpretable models (here linear models $g(z') = w_g \cdot z'$); D is a distance function; σ is the kernel width.

The roles of the two components asked about in the assignment:

- $\pi_x(z)$ defines "local": the closer a perturbed sample is to x, the more weight it gets. σ sets the neighbourhood size: a small σ leaves few samples with meaningful weight and makes results noisier, a large σ makes the surrogate lose its locality.
- $\Omega(g)$ penalises the complexity of the explanation. For linear models it is the number of non-zero weights, and the K-LASSO algorithm picks K features to keep the explanation short. This is the trade-off between local fidelity and readability.

Implementation note: the `lime` library for tabular data discretises continuous features by default, so a LIME weight is the effect of the value range the instance falls into, not a contribution relative to a baseline as in SHAP.

2.3 SHAP (Lundberg & Lee, 2017)

The paper defines the class of *additive feature attribution* methods: the explanation model is a linear function of binary variables,

$$g(z') = \varphi_0 + \sum_{j=1}^{M} \varphi_j z'_j, \qquad z' \in \{0,1\}^M \tag{3}$$

and proves that exactly one solution satisfies three properties (Theorem 1): *local accuracy* ($g(x') = f(x)$), *missingness* (an absent feature has no attributed effect) and *consistency* (if a model changes so that a feature's marginal contribution does not decrease, its attribution does not decrease). That solution is the Shapley value:

$$\varphi_j = \sum_{S \subseteq F \setminus \{j\}} \frac{|S|!\,(M - |S| - 1)!}{M!} \Big[f_{S\cup\{j\}}(x_{S\cup\{j\}}) - f_S(x_S) \Big] \tag{4}$$

Here F is the set of M features, S a subset not containing j, and f_S the prediction when only the features in S are known (the others are averaged out over background data). Three notions to remember:

- **Baseline** $\varphi_0 = \mathbb{E}[f(X)]$ is the average prediction over the background data, so it depends on how the background is chosen.
- **Contribution** φ_j is the part of the gap between $f(x)$ and the baseline that feature j accounts for; $\varphi_0 + \sum_j \varphi_j$ equals $f(x)$ exactly.
- **Sign** of φ_j: positive means the feature pushes the prediction above the baseline, negative means below.

Kernel SHAP estimates Shapley values by weighted linear regression, i.e. LIME with a specially chosen kernel, loss and regulariser (Theorem 2). The paper also notes that LIME's heuristic choices can violate local accuracy or consistency, and that estimation usually assumes feature independence (and, in some variants, model linearity). For a linear model with independent features, $\varphi_j = w_j(x_j - \mathbb{E}[X_j])$; this is the basis of the worked example in Section 3.2.

2.4 The survey by Mersha et al. (2024)

The survey classifies XAI techniques along four axes: scope (local/global), stage (ante-hoc/post-hoc), applicability (model-specific/model-agnostic) and design methodology (perturbation-based/gradient-based). For evaluating explanations it separates human-centred from computer-centred approaches and lists five computer-centred criteria: fidelity, consistency, robustness, efficiency and sufficiency, while stressing that no single metric suits every system. We use these criteria to organise the evaluation in Section 4.

2.5 Comparing the three sources

Table 2: Comparison of the three main sources.

Source	Goal	Main contribution	Limits / notes
Ribeiro et al. (2016)	Explain any classifier's prediction so humans can decide whether to trust it	LIME (sparse, weighted local surrogate) and SP-LIME for picking representative instances	Kernel and σ are heuristic; no Shapley-style guarantees
Lundberg & Lee (2017)	Unify feature-attribution methods	Additive attribution class; unique solution (Shapley values); Kernel SHAP	Estimation needs assumptions (feature independence); results depend on background data
Mersha et al. (2024)	Survey of XAI: taxonomy, applications, evaluation	Four-axis taxonomy; five evaluation criteria	Short on LIME/SHAP; some evaluation formulas need checking before use

The papers give concrete formulas and algorithms; the survey places them in the wider picture and names evaluation criteria.

3 Method and Worked Examples

3.1 Experimental setup

All choices below were fixed before looking at results, with fixed seeds so the experiment can be reproduced.

Table 3: Experimental setup.

Component	Choice	Reason
Data	Breast Cancer Wisconsin (scikit-learn): 569 instances, 30 features, class 0 = malignant, class 1 = benign; 80/20 stratified split, `random_state = 42` (455 train, 114 test)	Public, binary, readily available
Models	Logistic Regression (with standardisation) and a 200-tree Random Forest; test accuracy 98.2% and 95.6%	A linear model as a reference, a more opaque model to expose instability
Explanation space	Probability $P(\text{malignant})$	The original LIME explains probabilities; the two methods become directly comparable. Consequence: the φ values add up on the probability scale, not on log-odds
SHAP	KernelSHAP, 100 background instances drawn from train, 2000 sampled coalitions	One tool for both models so runtimes are comparable
LIME	5000 perturbed samples, default kernel width $0.75\sqrt{d} \approx 4.11$, default discretisation	Keeps the library's common configuration
Three instances	Fixed rule applied to the Random Forest: the correct prediction with the highest confidence (index 0), the smallest $\lvert P-0.5\rvert$ (index 16), the first misclassified instance (index 3)	Avoids cherry-picking "nice" examples
Number of runs	LIME: 20 seeds; SHAP: 5 background sets (seeds 0–4); top-k with $k = 5$	Enough to measure stability while staying fast
Deletion test	20 random test instances (fixed seed), up to 10 features removed by replacing them with the train mean	Checks faithfulness

The three instances:

Table 4: The three chosen instances.

Instance	Test index	True label	P(malignant) RF / LR
Confidently correct	0	malignant	1.000 / 1.000
Borderline	16	benign	0.495 / 0.634 (LR is wrong)
Misclassified	3	benign	0.745 (RF is wrong) / 0.466

3.2 Worked example 1: SHAP on a linear model

Assume the model $f(x) = b + \sum_j w_j x_j$, independent features, and the background is the mean $\mu_j = \mathbb{E}[X_j]$. Then $\varphi_0 = f(\mu)$ and $\varphi_j = w_j(x_j - \mu_j)$. Take $b = 1$, $w = (2, -1, 0.5)$, $\mu = (1, 2, 4)$ and the instance $x = (3, 1, 6)$.

- Baseline: $\varphi_0 = f(\mu) = 1 + 2 \cdot 1 - 1 \cdot 2 + 0.5 \cdot 4 = 3$.
- Prediction: $f(x) = 1 + 2 \cdot 3 - 1 \cdot 1 + 0.5 \cdot 6 = 9$.
- Contributions: $\varphi_1 = 2(3 - 1) = 4$; $\varphi_2 = (-1)(1 - 2) = 1$; $\varphi_3 = 0.5(6 - 4) = 1$.
- Check: $\varphi_0 + \varphi_1 + \varphi_2 + \varphi_3 = 3 + 4 + 1 + 1 = 9 = f(x)$.

Enumerating every subset with the original Shapley formula (with $v(S) = f$ where features in S take their value from x and the rest take the mean) gives the same result $(4, 1, 1)$. Feature 2 has $w_2 < 0$

but $x_2 < \mu_2$, so its contribution is still positive: the sign of φ_j depends on both the coefficient and where the instance lies relative to the background.

3.3 Worked example 2: LIME step by step

Take the small non-linear model $f(x) = \text{sigmoid}(x_1^2 - x_2)$ and the instance $x_0 = (1, 2)$, so $f(x_0) = \text{sigmoid}(-1) = 0.2689$. For traceability we use 6 fixed perturbed points instead of random sampling, $\sigma = 1$, and drop Ω (no sparsity penalty).

- **Perturb and query:** compute $f(z)$ at the 6 points z. For $z = (1.3, 2.2)$: $1.3^2 - 2.2 = -0.51$, so $f(z) = \text{sigmoid}(-0.51) = 0.3752$.
- **Distance and weight:** $D = \|z - x_0\|$, $\pi = \exp(-D^2/\sigma^2)$.
- **Weighted regression:** solve $\beta = (A^\top W A)^{-1} A^\top W y$ with $A = [\mathbf{1}, z - x_0]$, $W = \text{diag}(\pi)$, $y = f(z)$.

Table 5: The six fixed perturbed points of the LIME example.

z_1	z_2	$f(z)$	D	π
1.3	2.2	0.3752	0.361	0.878
0.6	2.1	0.1493	0.412	0.844
1.2	1.5	0.4850	0.539	0.748
0.9	1.7	0.2911	0.316	0.905
1.5	2.4	0.4626	0.640	0.664
0.7	2.5	0.1182	0.583	0.712

Result ($\sigma = 1$): the LIME coefficients are $g_1 = 0.395$ and $g_2 = -0.149$. The true derivatives of f at x_0 are $\partial f/\partial x_1 = f(1-f) \cdot 2x_1 = 0.393$ and $\partial f/\partial x_2 = -f(1-f) = -0.197$. So g_1 matches closely, while g_2 has the right sign but is off by about 0.05, because there are only 6 points and f curves within the neighbourhood; this illustrates that an explanation depends on the perturbation sample. Changing σ to 0.3 gives $(g_1, g_2) = (0.362, -0.132)$ and $\sigma = 3$ gives $(0.397, -0.150)$: in this tiny example σ matters little, but on the 30-dimensional data of Section 4 the sensitivity is measured directly.

4 Results and Discussion

4.1 Feature-contribution charts

Figure 1 (Random Forest) and Figure 2 (Logistic Regression) show the contributions of the 8 largest features. Orange pushes $P(\text{malignant})$ up, blue pushes it down. For SHAP, $\varphi_0 + \sum_j \varphi_j$ equals $f(x)$ exactly in all six cases (for example Random Forest, instance 0: $0.326 + 0.674 = 1.000$), as local accuracy requires.

- **Confidently correct instance (RF):** LIME and SHAP both put the tumour-size features (worst area, worst perimeter, worst radius, worst concave points) first, all pushing towards malignant.
- **Borderline instance (RF, $P = 0.495$):** contributions are small and mixed in sign; LIME puts worst concavity, mean texture and worst texture first, while SHAP puts worst perimeter, worst concavity and mean concave points (negative).
- **Misclassified instance (RF, $P = 0.745$):** see Section 4.3.

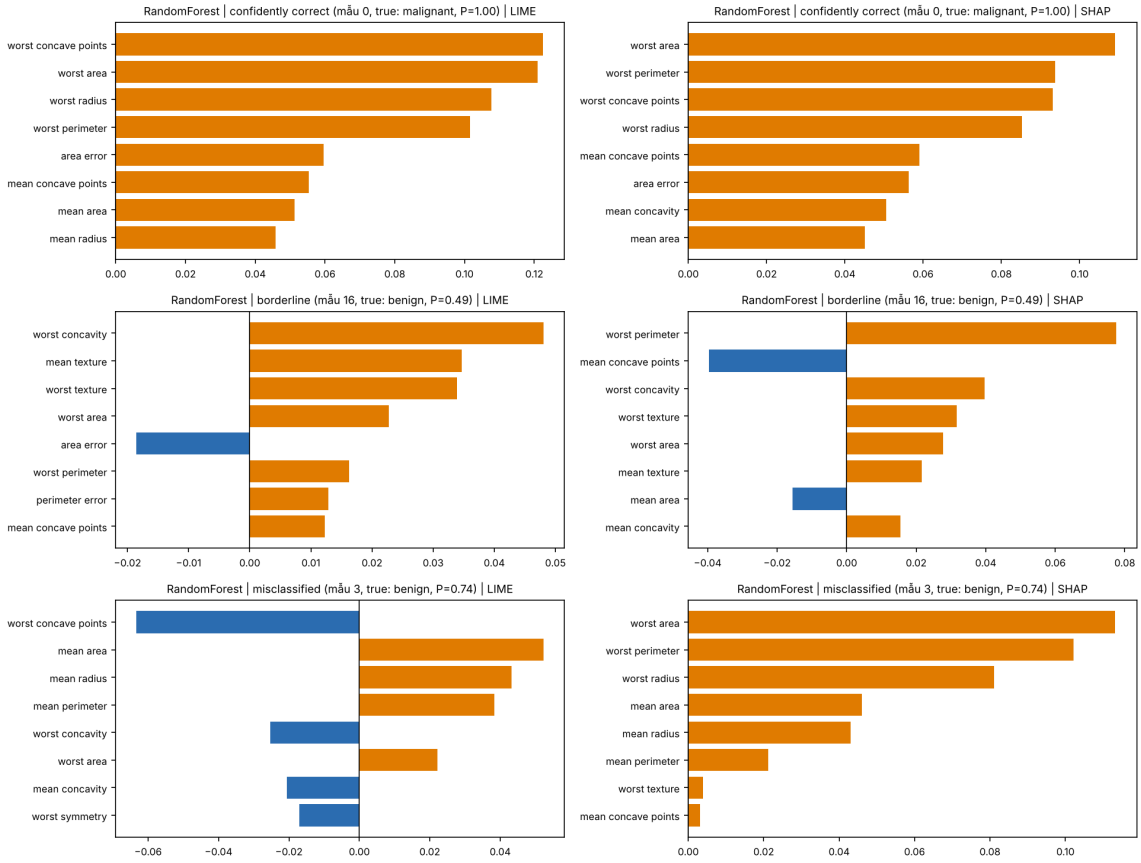


Figure 1: Feature contributions from LIME (left column) and SHAP (right column) for Random Forest; the three instances are, top to bottom, confidently correct, borderline and misclassified. Source: the team's experiment.

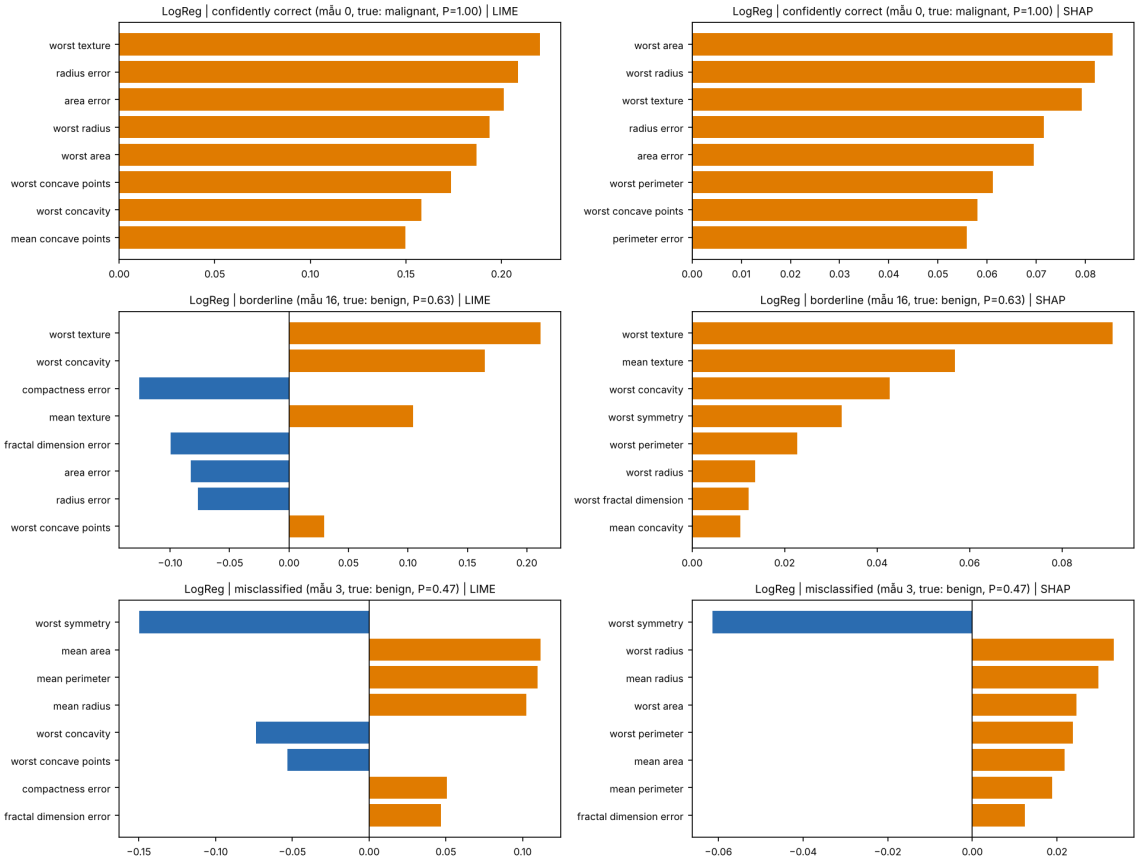


Figure 2: Same as Figure 1 for Logistic Regression. On instance 3 Logistic Regression predicts correctly ($P = 0.466$). Source: the team's experiment.

4.2 LIME versus SHAP

"Overlap@5" is the number of shared features in the two top-5 sets divided by 5, shown as (main run / average over all run pairs). Spearman is computed on |contribution| over all 30 features.

Table 6: LIME versus SHAP on the three instances.

Model	Instance	Overlap@5	Spearman	Top-1 LIME	Top-1 SHAP	Runtime L / S (s)
LogReg	Confidently correct	1.0 / 0.80	0.74	worst texture	worst area	0.034 / 0.036
LogReg	Borderline	0.6 / 0.67	0.24	worst texture	worst texture	0.032 / 0.031
LogReg	Misclassified	0.4 / 0.49	0.26	worst symmetry	worst symmetry	0.033 / 0.030
Random Forest	Confidently correct	0.8 / 0.82	0.83	worst concave points	worst area	0.064 / 0.389
Random Forest	Borderline	0.6 / 0.71	0.69	worst concavity	worst perimeter	0.066 / 0.451
Random Forest	Misclassified	0.4 / 0.44	0.46	worst concave points	worst area	0.062 / 0.391

Overlap@5 falls as the instance gets harder, for both models: from 0.8–1.0 on the confidently correct instance to 0.4 on the misclassified one. When both methods put a feature in the top-5, the sign always agrees (sign agreement 1.0 in all six cases); the disagreement is about *which* features are chosen, not about the direction of the effect. Spearman is lowest for Logistic Regression (0.24–0.26 on the two hard instances): the ranking of all 30 features differs a lot even though this is the simplest model. One possible reason is that LIME discretises features, but we did not test this. On runtime,

the methods are similar on Logistic Regression (about 0.03 s), but KernelSHAP is about 6–7 times slower than LIME on Random Forest (0.39–0.45 s vs. 0.06–0.07 s) because it queries the model on 2000 coalitions × 100 background instances; timings come from one machine.

4.3 An example where explanations differ: instance 3 (Random Forest is wrong)

For the same prediction, $P(\text{malignant}) = 0.745$ (the true label is benign), the two methods tell different stories:

Table 7: Top-5 features of LIME and SHAP on instance 3 (Random Forest).

Method	Top-5 by |contribution|	Signs
LIME	worst concave points, mean area, mean radius, mean perimeter, worst concavity	worst concave points and worst concavity negative; the rest positive
SHAP	worst area, worst perimeter, worst radius, mean area, mean radius	All positive

Only two features are shared (mean area, mean radius), i.e. overlap@5 = 0.4. SHAP explains the prediction almost entirely through size features that all push towards malignant; LIME gives a mixed picture whose largest feature is worst concave points with a negative sign. There is no ground truth to say which is right. One reasonable reading is that the two explanations answer slightly different questions (LIME: within the neighbourhood of the instance, which value ranges matter; SHAP: contribution relative to the average baseline), and this very gap is why faithfulness must be checked (Section 4.8).

4.4 Stability

"Jaccard" is the mean similarity of the top-5 sets over all pairs of runs (1 = perfectly stable); LIME varies the seed (20 runs), SHAP varies the background set (5 runs). "Top-1 dominance" is the share of runs in which the top feature is the one most often ranked first.

Table 8: Stability of the top-5.

Model	Instance	Jaccard LIME	Top-1 LIME	Jaccard SHAP	Top-1 SHAP
LogReg	Confidently correct	0.78	0.95	0.68	0.40
LogReg	Borderline	0.84	1.00	0.73	1.00
LogReg	Misclassified	0.85	1.00	0.53	1.00
Random Forest	Confidently correct	0.94	0.60	1.00	1.00
Random Forest	Borderline	0.72	0.80	0.73	1.00
Random Forest	Misclassified	0.74	1.00	1.00	1.00

The top-5 Jaccard ranges from 0.72 to 0.94 for LIME and from 0.53 to 1.00 for SHAP. SHAP is perfectly stable (1.00) on Random Forest for the confidently correct and misclassified instances, yet it has the lowest value in the table (0.53) on Logistic Regression, instance 3. Neither method is consistently more stable. Top-1 dominance is low in a few cells (LIME 0.60 on RF instance 0; SHAP 0.40 on LR instance 0): when P is near 1 many features contribute about equally, so the top rank flips easily; this is a proposed explanation that we did not test separately.

An unstable example (Random Forest, borderline instance, Figure 3). The three features worst concavity, worst texture and worst perimeter are almost always in the top-5 of both methods. The difference is at ranks 4–5: LIME picks mean texture in 100% of runs and area error in 45%,

whereas SHAP picks mean concave points in 100% of runs (LIME picks it in only 10%) and mean texture in just 40%. LIME's top feature also changes: seed 0 ranks worst concavity first, seeds 1–5 rank worst texture first.

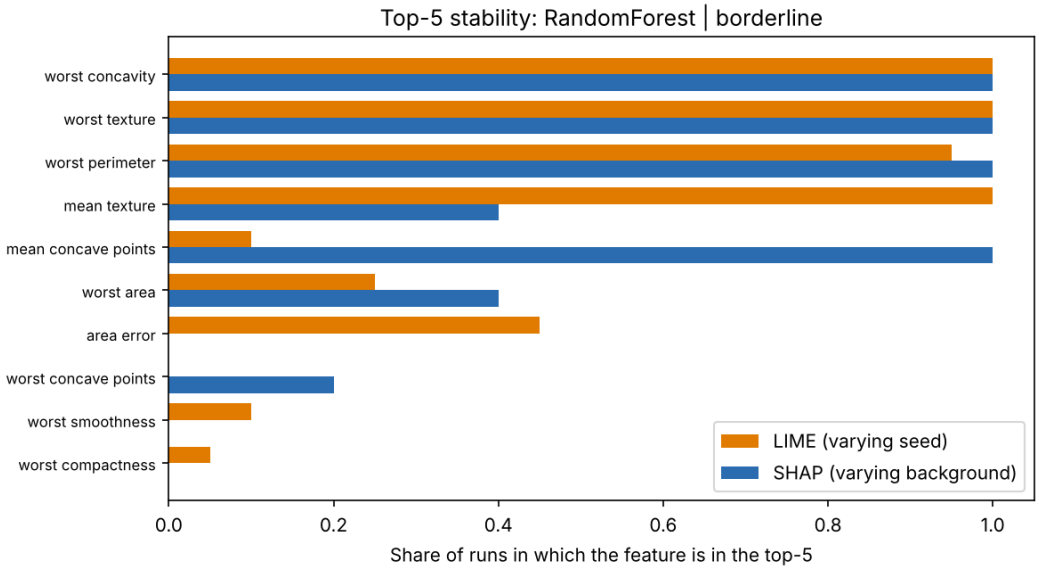


Figure 3: Share of runs in which each feature is in the top-5, Random Forest, borderline instance (LIME: 20 seeds; SHAP: 5 background sets). Source: the team's experiment.

4.5 Sensitivity to background size and kernel width

Tested on Random Forest, borderline instance (SHAP: 5 background sets per size; LIME: 10 seeds per kernel width, fewer than the 20 seeds of Section 4.4, so the Jaccard values are not directly comparable).

Table 9: Sensitivity to parameters.

Method	Parameter	Jaccard top-5	Top-1 dominance
SHAP	background = 10	0.63	0.8
SHAP	background = 50	0.73	1.0
SHAP	background = 100	0.73	1.0
LIME	kernel width = 1.00	0.65	0.7
LIME	kernel width = 2.00	0.68	0.9
LIME	kernel width = 4.11 (default)	0.68	0.9
LIME	kernel width = 8.00	0.68	0.9

A background of only 10 instances makes SHAP less stable, but from 50 on the Jaccard does not rise further (0.73). For LIME, a small kernel width (1.0) is slightly less stable and values from 2.0 up make no difference. The best Jaccard values (0.73 and 0.68) are still below 1, so changing the parameters does not remove the instability on this instance.

4.6 Checking faithfulness with a deletion test

On 20 random test instances, we remove up to 10 of the top-ranked explained features in the direction of the predicted class (replacing them with the train mean) and track the probability of that class. The baseline is a random order (average of 10 permutations).

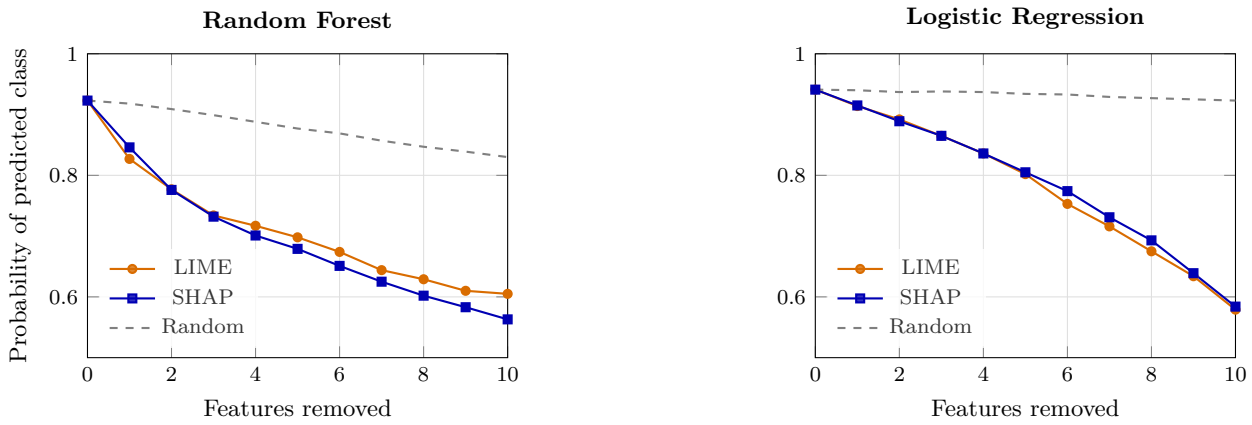


Figure 4: Deletion test, 20 test instances, one seed. Values after 5 features are removed: Random Forest 0.698 (LIME), 0.679 (SHAP), 0.877 (random); Logistic Regression 0.802, 0.805, 0.934. Source: the team's `results/deletion_curves.csv`.

Both LIME and SHAP lower the probability much faster than random removal. After 5 features are removed, the average probability of Random Forest falls from 0.923 to 0.698 (LIME) and 0.679 (SHAP), while random removal only reaches 0.877; for Logistic Regression the values are 0.802 (LIME), 0.805 (SHAP) and 0.934 (random, starting from 0.941). At 5 features LIME and SHAP differ by no more than 0.02; at 10, on Random Forest, SHAP is lower than LIME (0.563 vs. 0.605). With one seed and 20 instances, and no confidence intervals, this small gap is not enough to say which method is more faithful. Caveats: replacing a feature with its mean can create out-of-distribution samples, and because the features are strongly correlated, removing one does not necessarily remove its information.

4.7 Feature importance is not a causal effect

LIME and SHAP describe how the *model* uses features, not how the features affect the *disease*. There are four reasons, each visible in our experiment:

- **Correlated features.** worst radius, worst perimeter and worst area are tightly linked geometrically (perimeter and area both depend on radius) and all appear at the top of the charts; how credit is split among them depends on the model and the background data, not on a separate cause.
- **Out-of-distribution samples.** LIME and KernelSHAP change feature values almost independently of each other (SHAP additionally assumes independence), creating combinations that do not occur, such as a large area with a small radius; the model is queried where it has never seen data.
- **Explanations reflect the model's mistakes too.** On instance 3 the size features "pushing towards malignant" tell us why the model erred, not that the tumour is malignant.
- **Real interventions differ from value replacement.** Removing or replacing a feature in the algorithm is a computation on the model; changing tumour size in the real world requires causal assumptions about the data-generating process that neither method tests.

Conclusions should therefore read "the model relies on X", not "X causes Y".

4.8 A procedure for evaluating faithfulness (proposed)

We propose a five-step procedure tied to the criteria of Mersha et al. (2024). We use those criteria in spirit and do not use the survey's exact formulas.

- **Sanity check on a model with a known answer (fidelity).** For a linear model the explanation must match $w_j(x_j - \mu_j)$. Done on the hand-worked example (Section 3.2); not yet compared against the Logistic Regression of the experiment.
- **Local fit (fidelity).** For LIME, check that the surrogate tracks f in the neighbourhood (for example a weighted R^2); for SHAP, check $\varphi_0 + \sum_j \varphi_j = f(x)$. SHAP additivity was checked (Section 4.1); the LIME R^2 was not computed.
- **Deletion/insertion versus random (fidelity, in the spirit of sufficiency).** Remove, or keep only, the top-ranked features and compare with a random order. The deletion test was run (Section 4.6); the insertion test was not.
- **Stability (consistency).** Repeat over many seeds and backgrounds and report top-k Jaccard and variance. Done (Sections 4.4–4.5).
- **Robustness.** Perturb the input slightly, or compare explanations across models of equal accuracy, and measure how much the explanation changes. Not done.

Efficiency is measured by runtime (Section 4.2). An explanation should be regarded as reliable only if it passes several of these steps together; passing one is not enough.

4.9 Limitations

- Only one dataset, three instances and two models; the results may not generalise.
- 20 seeds (LIME) and 5 background sets (SHAP) are few, and there are no confidence intervals.
- The deletion test uses one seed and 20 instances, and replacing with the mean can create out-of-distribution samples.
- Runtimes are from one machine and for KernelSHAP; TreeSHAP for Random Forest would be much faster (not measured).
- Explanations are on the probability scale; on log-odds the results may differ (not tested).
- LIME discretises features, so only ranks and signs can be compared with SHAP, not absolute magnitudes.
- There is no ground truth for the Random Forest's explanations, so we cannot say which method is "correct".

5 Conclusion

No: high accuracy does not necessarily come with a reliable explanation. The two models reach 98.2% and 95.6% accuracy, yet the top-5 sets of LIME and SHAP overlap by only 0.4 on the misclassified instance, and each method's top-5 changes with the seed or background, with Jaccard as low as 0.53–0.72 on some instances. Both methods are more faithful than random removal in the deletion test, so they are not useless, but that test does not separate the two methods, and neither proves a causal relationship.

Conditions of use. Read an explanation as a description of the model's behaviour around one instance (LIME) or relative to a baseline (SHAP); report the seed, background data and kernel width; repeat the explanation and report its stability; add the faithfulness checks of Section 4.8, especially for borderline and misclassified instances, where explanations are most often used to support decisions.

Limitations. One dataset, three instances, two models, few repetitions and no confidence intervals (Section 4.9), so the conclusions are illustrative.

Next steps. Run the insertion test and the LIME R^2, try TreeSHAP and log-odds, add datasets and models, and increase the number of repetitions to obtain confidence intervals.

References

[1] Lundberg, S. M., & Lee, S.-I. (2017). A Unified Approach to Interpreting Model Predictions. *NeurIPS 2017*. Paper page.

[2] Mersha, M., Lam, K., Wood, J., AlShami, A., & Kalita, J. (2024). Explainable artificial intelligence: A survey of needs, techniques, applications, and future direction. *Neurocomputing*, 599, 128111. DOI (we read the arXiv version 2409.00265v2).

[3] Ribeiro, M. T., Singh, S., & Guestrin, C. (2016). "Why Should I Trust You?": Explaining the Predictions of Any Classifier. *KDD 2016*, 1135–1144. DOI.

Data, tools and source of the numbers

- Data: Breast Cancer Wisconsin (Diagnostic), loaded through `sklearn.datasets.load_breast_cancer`.
- Libraries: scikit-learn, lime, shap (Python).
- All figures, tables, numbers and the two worked examples in this report were run or computed by the team with the notebook `S04_LIME_SHAP_experiments.ipynb`; none are taken from the papers.
- AI tools: Claude (Anthropic) was used to help explain concepts, write the experiment code and draft this report.